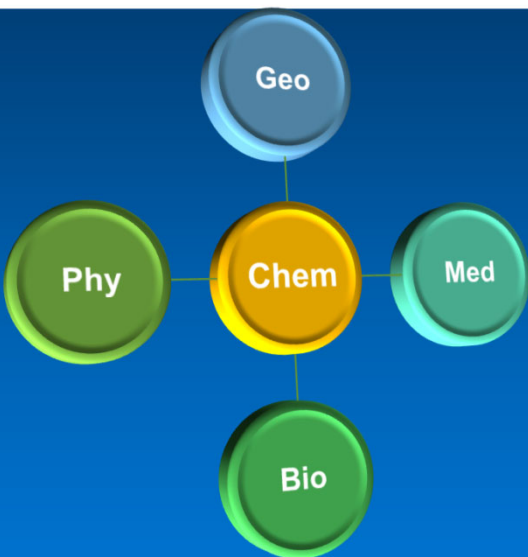


Fundamental Chemistry I

酸碱理论



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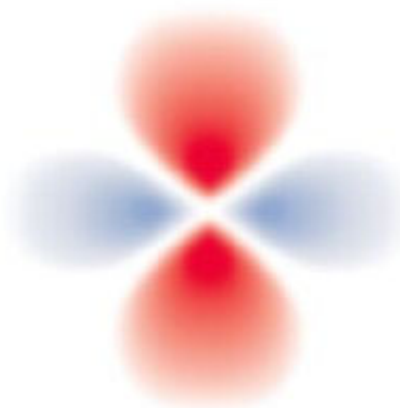




second edition

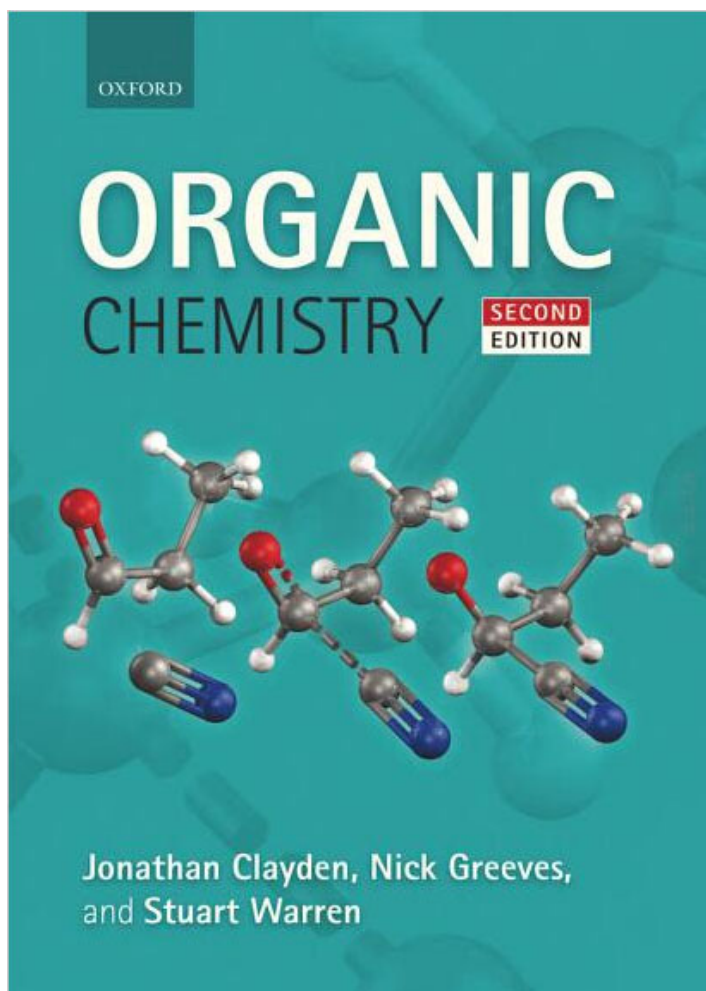
CHEMICAL STRUCTURE and REACTIVITY

an integrated approach



James Keeler | Peter Wothers

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- pH与pKa
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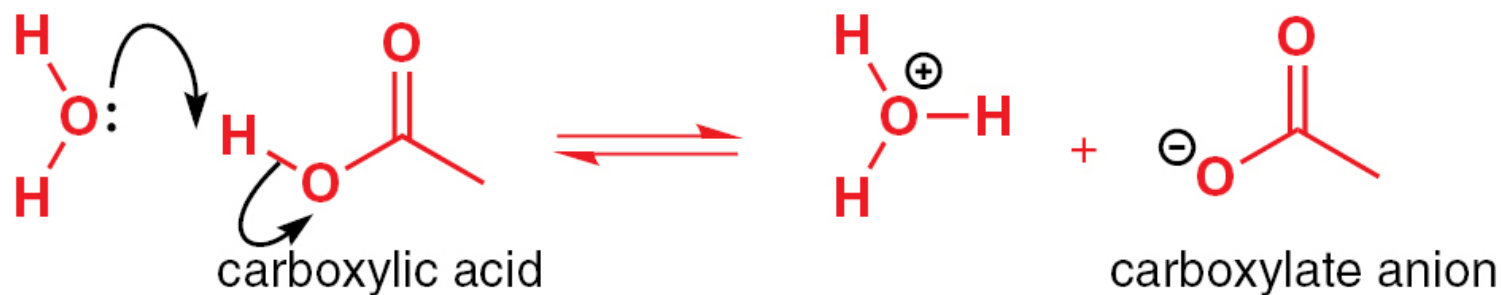
酸碱反应

- 酸碱反应是非常基础的化学反应。绝大多数的化学反应要么本身是酸碱反应，要么在反应的过程中包含了酸碱反应。
- 通过学习酸碱反应，可以学习化学家是如何使用弯箭头来表示机理以及如何描述分子反应过程中化学键的断裂和生成。



Brønsted–Lowry酸碱理论

- Brønsted–Lowry酸碱反应包含了质子转移。
- Brønsted–Lowry酸是指能够提供质子的物质。
- Brønsted–Lowry碱是指能够接受质子的物质。



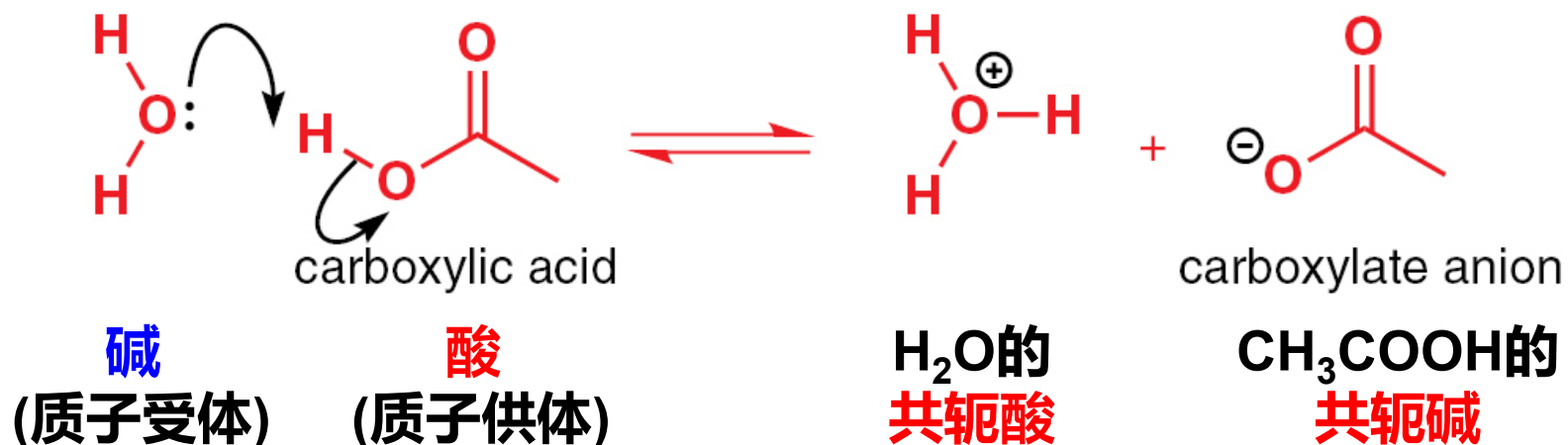


共轭酸碱

● For any acid and any base:



AH is an acid and A^- is its conjugate base and B is a base and BH^+ is its conjugate acid. That is, every acid has a conjugate base associated with it and every base has a conjugate acid associated with it.



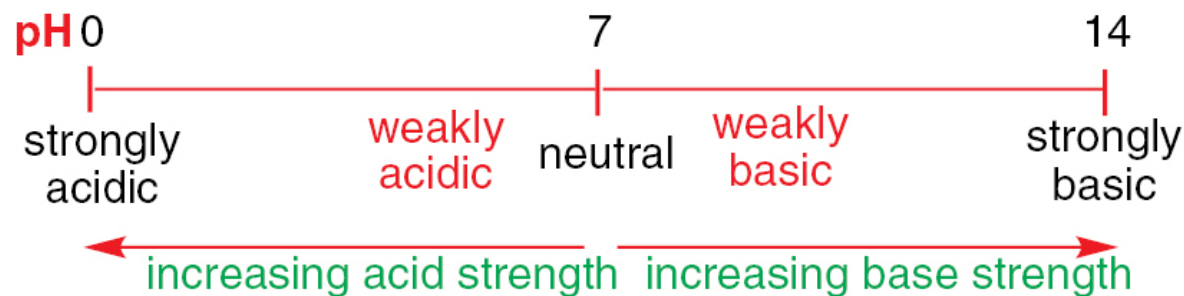
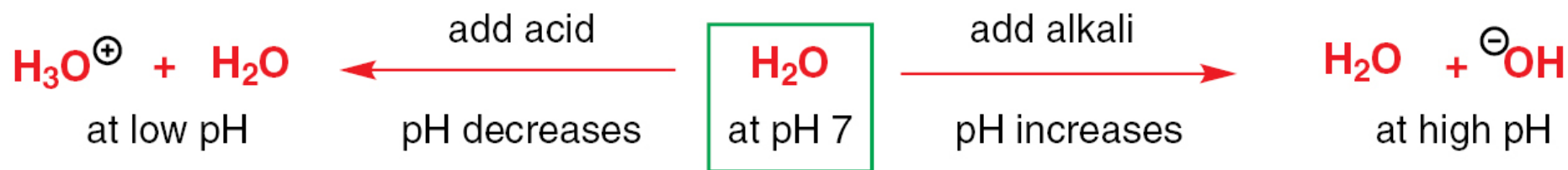
H_2O 既可以作为酸也可以作为碱.



pH范围

pH定义为 H_3O^+ 浓度对数的负值:

$$\text{pH} = -\lg[\text{H}_3\text{O}^+]$$



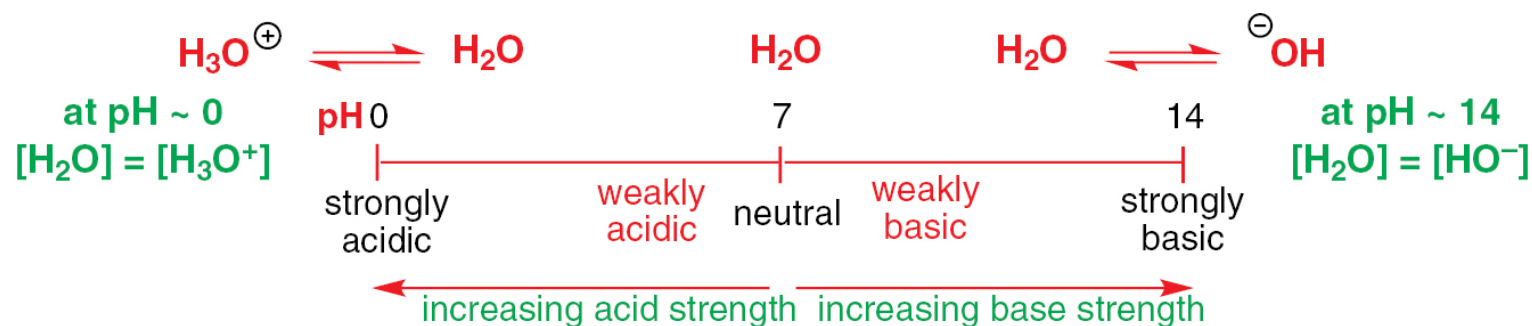


水的离子化



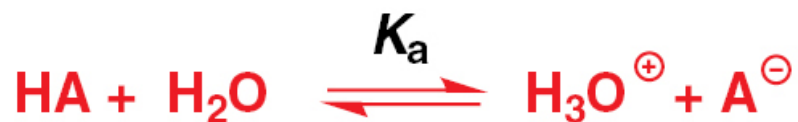
$$K_w = 1.01 \times 10^{-14} = [\text{H}_3\text{O}^+][\text{OH}^-]$$

K_w 随着温度改变而改变，例如0 °C时是 1.15×10^{-15} ，100 °C时是 4.99×10^{-13} 。





pKa的定义



$$K_a = \frac{[\text{H}_3\text{O}^{\oplus}] [\text{A}^{\ominus}]}{[\text{AH}]}$$

$$\text{p}K_a = -\log K_a$$

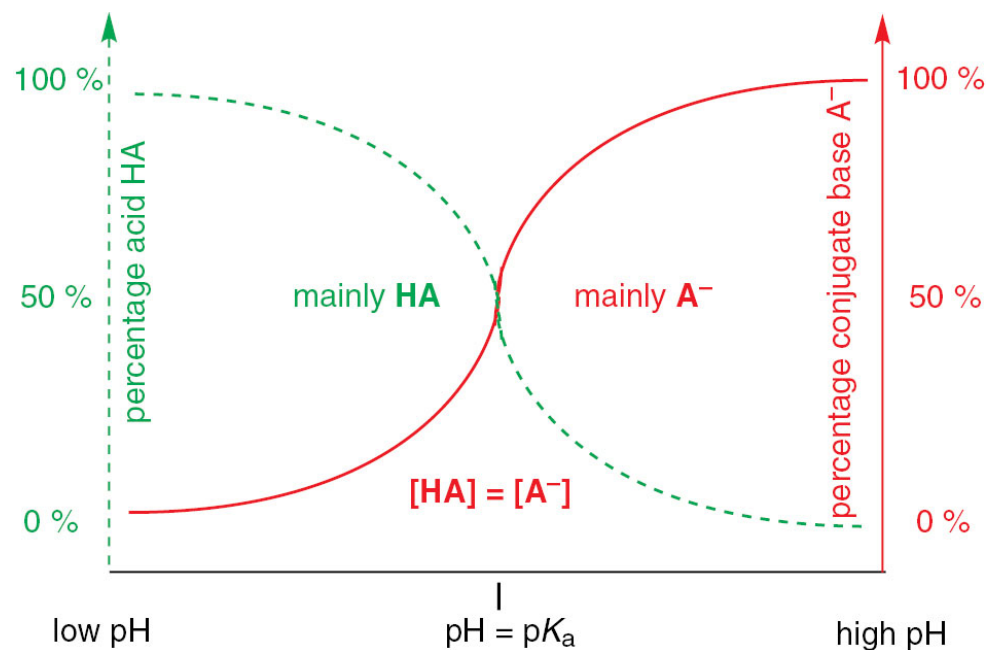
pKa可以用来预测某一溶液中的主要物种:



$$K_a = 10^7$$



$$K_a = 1.74 \times 10^{-5}$$





pKa的用途

pKa可以用来比较酸性和碱性：

- Acid and conjugate base strength

The stronger the acid HA, the weaker its conjugate base A⁻.
The stronger the base A⁻, the weaker its conjugate acid AH.

The approximate pK_a values of some inorganic compounds

Acid	pK _a	Conjugate base	Acid	pK _a	Conjugate base	Acid	pK _a	Conjugate base
H ₂ SO ₄	-3	HSO ₄ ⁻	H ₃ O ⁺	-1.7	H ₂ O	NH ₄ ⁺	9.2	NH ₃
HCl	-7	Cl ⁻	H ₂ O	15.7	HO ⁻	NH ₃	33	NH ₂ ⁻
HI	-10	I ⁻	H ₂ S	7.0	HS ⁻			

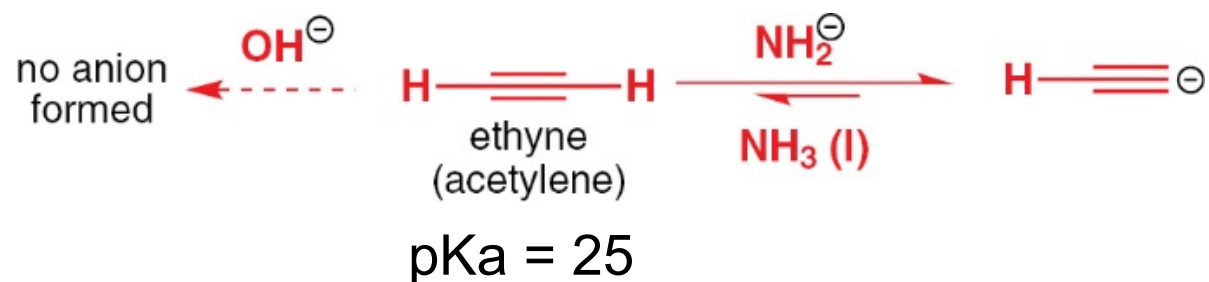
pKa会随溶剂的变化而变化！



pKa与溶剂

溶剂选择会限制我们能利用的pKa范围。

在某种溶剂中，我们能利用的**最强酸**是溶剂分子的**共轭酸**，而我们能利用的**最强碱**是溶剂分子的**共轭碱**。这种效应被成为“**拉平效应**”。例如，在水中，最强的酸是 H_3O^+ 而最强的碱是 OH^- 。因此，我们不能在水中比较 HCl 和 HBr 的酸性 (均强于 H_3O^+) 或者 NH_2^- 和 CH_3^- 。





影响pKa的因素

一系列的因素会影响AH的酸性:

- 共轭碱的**稳定性**。这是影响酸性最重要的因素。影响共轭碱稳定性的一个重要因素是负电荷所在的元素。负电荷所在的元素电负性越高，共轭碱就越稳定。
- A—H键的**强度**。键越容易断裂，酸性越强。
- 溶剂**稳定离子的能力**。溶剂稳定离子的能力越强，在该溶剂中的酸性就越强。



影响pKa的因素

高电负性元素上的负电荷能够稳定共轭碱:

Acid	Conjugate base	pK _a
methane CH ₄	CH ₃ ⁻	~48
ammonia NH ₃	amide ion NH ₂ ⁻	~33
water H ₂ O	hydroxide ion HO ⁻	~16
HF	fluoride ion F ⁻	3

弱的A-H键造就强酸:

Acid	Conjugate base	pK _a
HF	fluoride ion F ⁻	3
HCl	chloride ion Cl ⁻	-7
HBr	bromide ion Br ⁻	-9
HI	iodide ion I ⁻	-10



影响pKa的因素

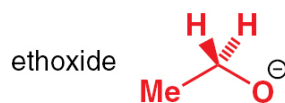
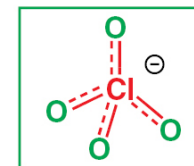
Acidity increases within a given row (electronegativity effect)					Acidity increases within a given column (bond strength effect)
Hydride pK_a	C (H ₃ C—H) 48	N (H ₂ N—H) 38	O (HO—H) 15.7	F (F—H) 3.2	
			S (HS—H) 7.0	Cl (Cl—H) -7	
			Se (HSe—H) 3.9	Br (Br—H) -9	
				I (I—H) -10	



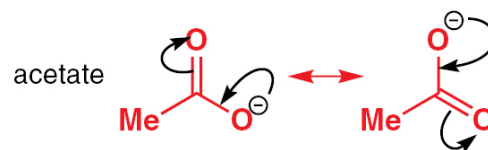
影响pKa的因素

负电荷的离域能够稳定共轭碱:

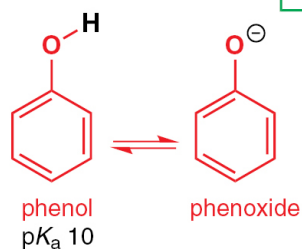
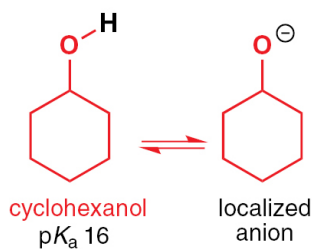
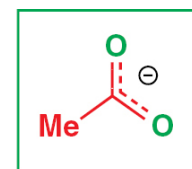
Acid	Conjugate base	pK _a
hypochlorous acid HO-Cl	ClO ⁻	7.5
chlorous acid HO-ClO	ClO ₂ ⁻	2
chloric acid HO-ClO ₂	ClO ₃ ⁻	-1
perchloric acid HO-ClO ₃	ClO ₄ ⁻	-10



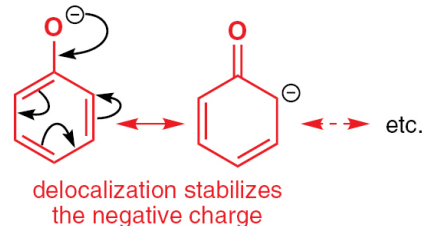
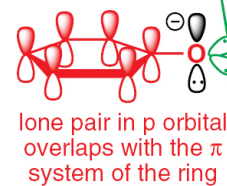
charge localized on one oxygen



charge delocalized over two oxygens



these lone pairs in sp² orbitals do not overlap with the π system of the ring

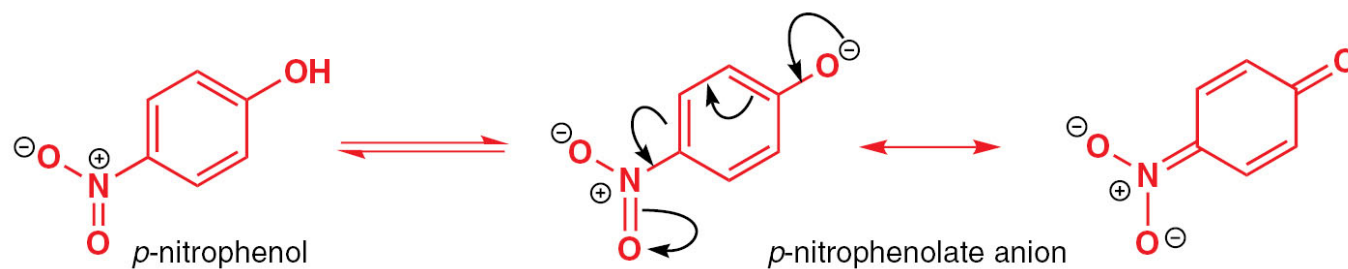




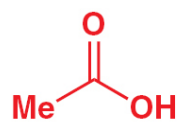
影响pKa的因素

能够与得失质子的位点（原子）产生共轭的取代基以及虽然没有共轭但是具有高电负性的取代基，均能对pKa产生显著影响。

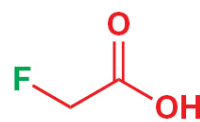
共轭效应:



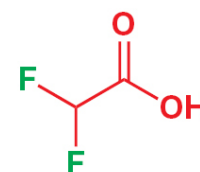
诱导效应:



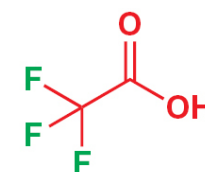
acetic acid
 pK_a 4.76



fluoroacetic acid
 pK_a 2.59



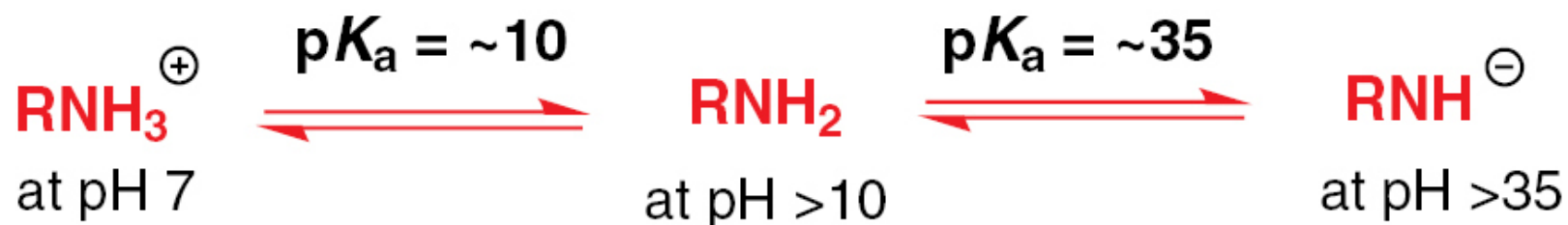
difluoroacetic acid
 pK_a 1.34



trifluoroacetic acid
 $pK_a \sim -1$



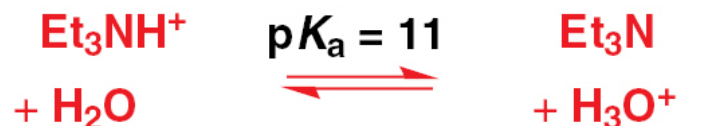
含氮化合物的酸和碱



碱的“pKa”:

the triethylammonium ion:

triethylamine's
conjugate acid

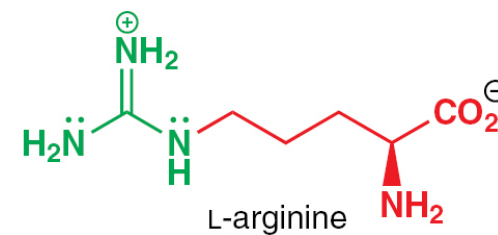
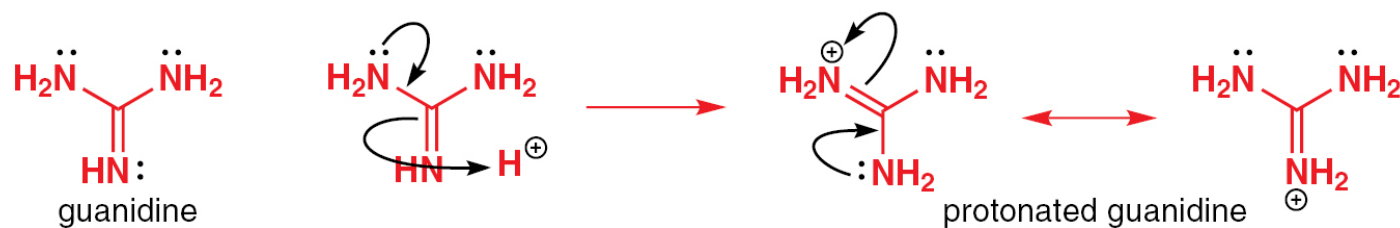
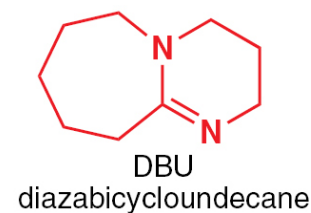
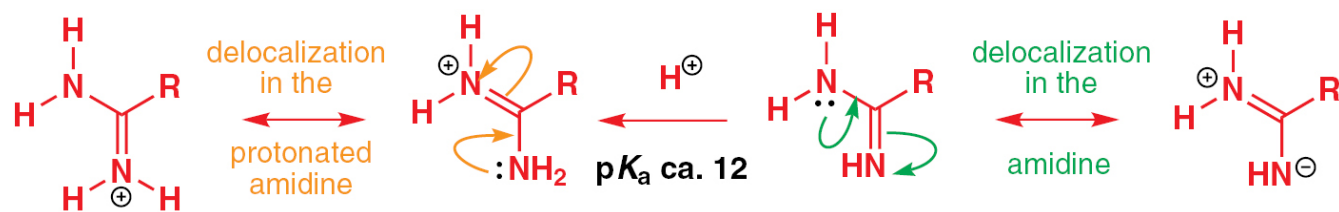
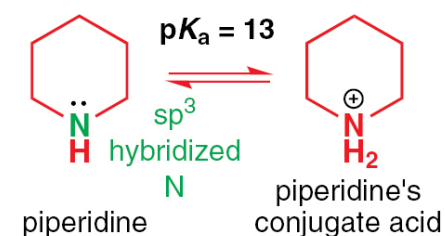
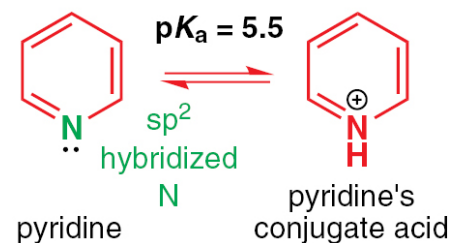
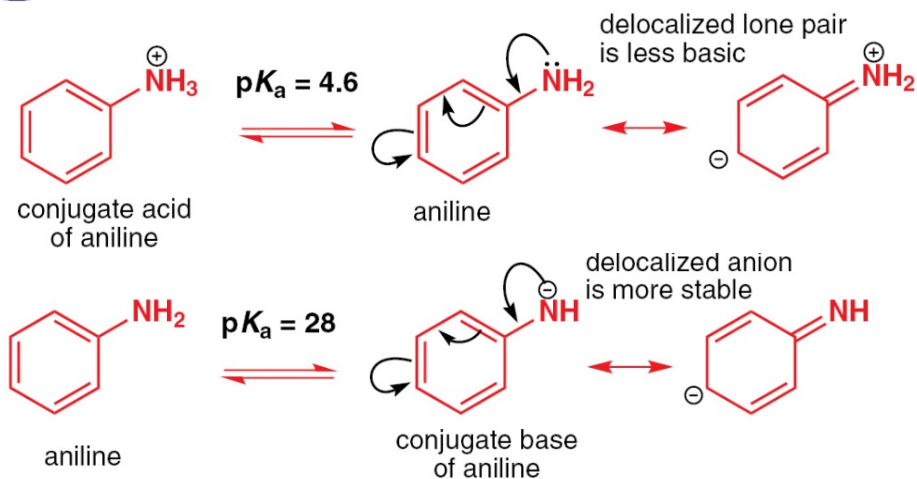


triethylamine

It's OK to say 'the $\text{p}K_a$ of triethylamine is about 10' as long as you understand that what is really meant is 'the $\text{p}K_a$ of the triethylammonium ion is about 10', which can also be expressed thus: 'the $\text{p}K_{\text{aH}}$ of triethylamine is about 10'

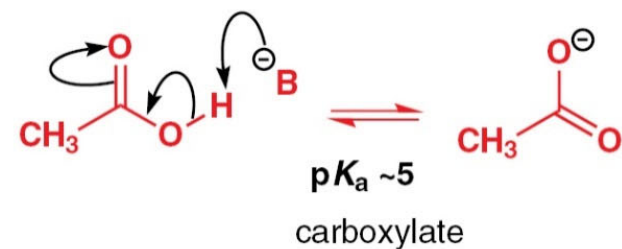
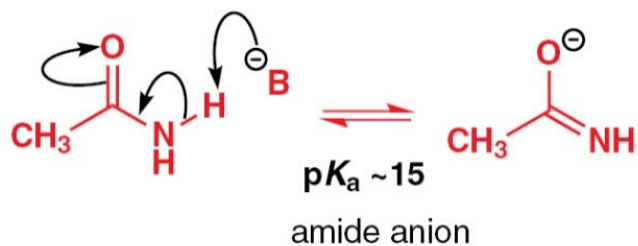
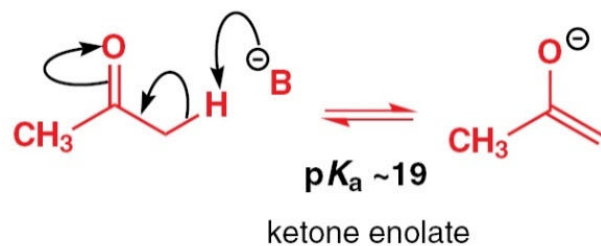
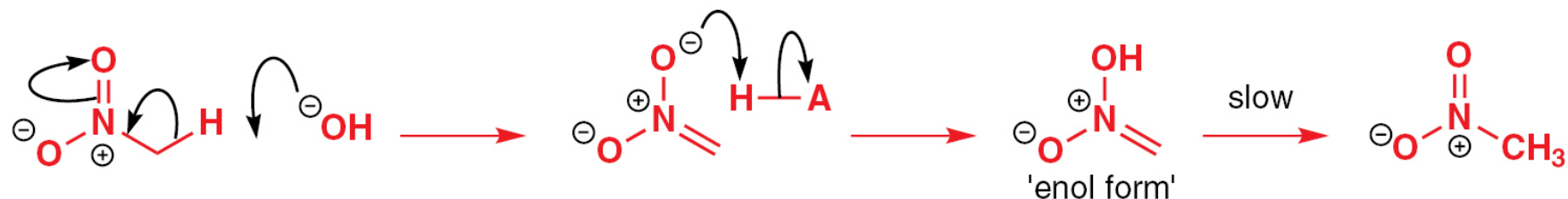
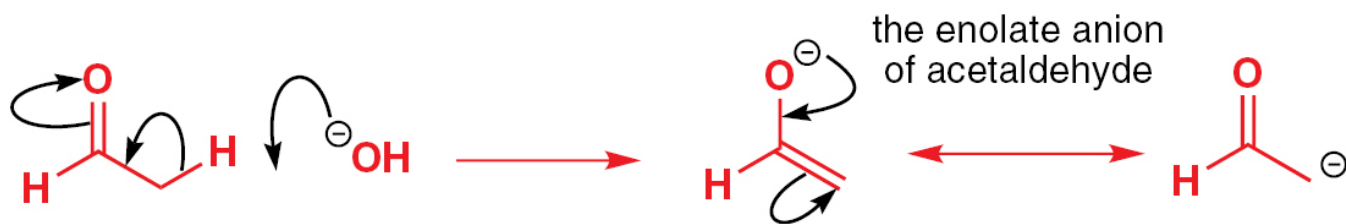


常见的含氮碱





常见的含碳酸

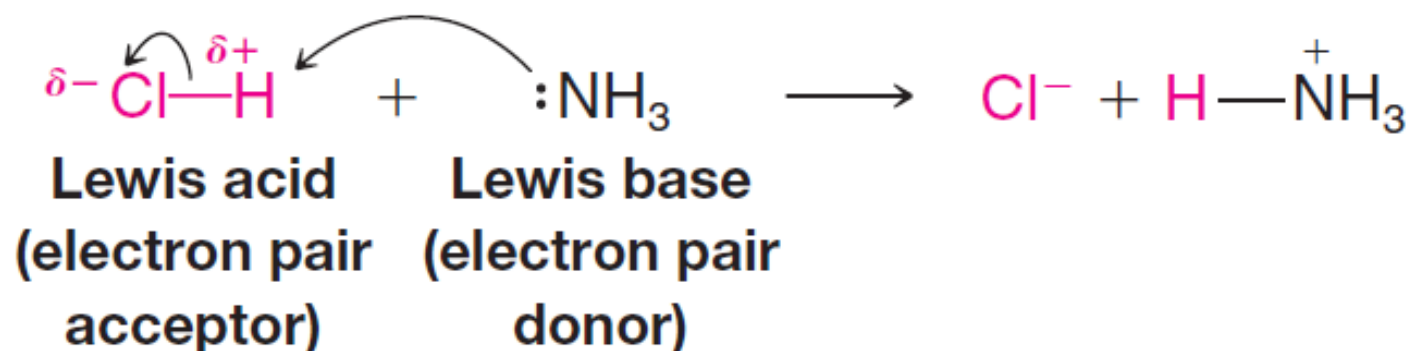




Lewis酸碱理论

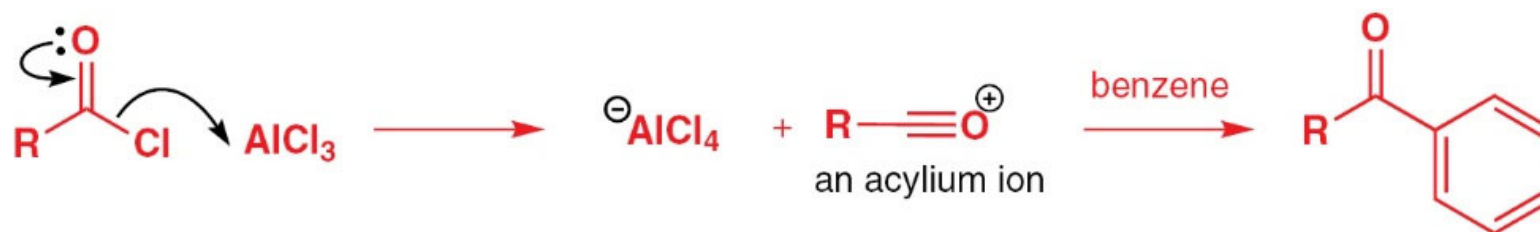
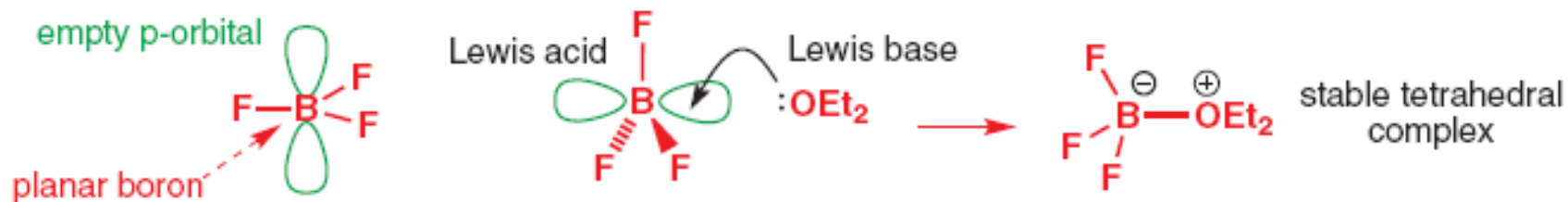
1923年, G. N. Lewis提出一个酸碱理论, 显著拓展了化学家对酸和碱的认识。

- Lewis酸是能够接受电子对的物质。
- Lewis碱是能够提供电子对的物质。





Lewis酸碱



Lewis酸通常和带有孤对电子的高电负性原子强烈作用，比如卤素和氧。



复习要点

- Brønsted–Lowry酸碱理论
- pH与pKa
- 影响pKa的因素
- Lewis酸碱理论



水的pKa——一点思考

$$K_a = \frac{[H^+][OH^-]}{[H_2O]_l}$$

$$[H_2O]_l = 1$$

$$pK_a = -\lg K_w = 14$$

$$K_a = \frac{[H^+][OH^-]}{[H_2O]}$$

$$[H_2O] = 55.6 \text{ mol/L}$$

$$pK_a = -\lg\left(\frac{K_w}{[H_2O]}\right) = 15.7$$

进行热力学的相关计算，需使用pKa = 14;
比较“独立”酸性（即不考虑溶剂），用pKa = 15.7比较合理。

pK_a Values in the Undergraduate Curriculum: What Is the Real pK_a of Water? 10.1021/acs.jchemed.6b00623
Confusing Quantitative Descriptions of *Brønsted-Lowry* Acid-Base Equilibria in Chemistry Textbooks – A Critical Review and Clarifications for Chemical Educators 10.1002/hlca.201300321